SP-30-9 Trajectory Spectroscopy at Limit Interfaces (v0.1)

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SP-30-9 Trajectory Spectroscopy at Limit Interfaces

Headline claim

TARGET-PREDICTION: At algebraic-transcendental limit interfaces, BST predicts a class of limit-undecidable trajectories with γ_EM trajectory signature (per Toy 1157 + memory_feedback_gamma_trajectory.md + memory_feedback_limits_lossy.md).

γ = Euler-Mascheroni constant is trajectory NOT number; its limit destroys trajectory's transcendence information. BST predicts: - Fold catastrophe (per Casey's catastrophe theory connection): at the boundary between algebraic + transcendental observable trajectories - γ_EM signature: precision spectroscopic transitions near algebraic/transcendental boundaries should exhibit substrate-natural BST primary modulation - Multi-domain testable: across atomic + nuclear + cosmological scales

Substrate-mechanism articulation

γ as limit-undecidable trajectory (Toy 1157):

The Euler-Mascheroni constant γ = lim_{n→∞} (H_n - ln n) ≈ 0.5772 emerges as a limit of a trajectory. The limit operation is lossy compression (per feedback_limits_lossy.md): destroys trajectory information including whether γ is algebraic or transcendental.

Casey's catastrophe theory connection (per feedback_gamma_trajectory.md): At the boundary between algebraic + transcendental observable trajectories, a fold catastrophe occurs in substrate observable structure. The fold separates: - Algebraic side (rational + algebraic numbers) - Transcendental side (e, π, γ candidate, …)

Substrate-natural γ_EM signature:

Per T2476 α^{BST primary} substrate-coordinate count + γ-trajectory framework:

precision measurements near algebraic/transcendental limits should show BST primary integer modulation at α^k orders.

Experimental concept

Test platforms:

1. Precision atomic spectroscopy near series limits: Rydberg states approaching ionization continuum (algebraic-transcendental interface)
2. Nuclear decay near branching ratios: branching at transcendental ratios
3. Cosmological observables at limit-interfaces: dark energy equation of state w = -1 (algebraic-transcendental boundary)

BST falsifier: at algebraic-transcendental limit interfaces, observable trajectories should exhibit: - BST primary integer modulation at α^k orders (T2476) - Fold-catastrophe-like structure (theoretical, requires K52a + catastrophe framework) - Distinguishable from null-model random-trajectory hypothesis

Experimental program

Cost: $50-100K (statistical analysis-focused + precision spectroscopy collaboration)

Components: - Rydberg-state spectroscopy access (~$20-30K, existing facilities) - Statistical trajectory analysis pipeline (~$10-20K) - Cosmological data re-analysis (~$10-20K) - Student researcher (~$10-30K)

Timeline: 12-24 months from setup to results

Falsifier protocol:

1. Identify limit-interface observables: Rydberg series limits + nuclear decay branching + cosmological w-parameter
2. High-precision trajectory analysis: extract trajectories approaching algebraic/transcendental boundaries
3. Search for BST primary modulation: α^k corrections to trajectory at limit
4. Statistical significance: 2-3σ detection of substrate-natural pattern → BST confirmed at this prediction
5. Outcome:
    - 2σ BST pattern → trajectory spectroscopy framework supported
    - No pattern → BST trajectory-spectroscopy hypothesis refuted (other predictions independent)

Falsifier sharpness: MEDIUM. Limit-interface measurements are accessible; statistical power depends on dataset size.

γ_EM specific prediction

Per Toy 1157 trajectory classification: BST predicts that observable trajectories near transcendental constants should exhibit a substrate-natural BST primary "memory" — a small modulation reflecting their α-order substrate-coupling history.

Specifically: in atomic Rydberg spectroscopy near series limit (E → ionization threshold), substrate framework predicts a ~1/N_max ≈ 0.73% modulation pattern distinguishable from baseline.

Cross-link to limits + catastrophe framework

- Toy 1157: γ trajectory classification (limit-undecidable)
- feedback_gamma_trajectory.md: γ as trajectory + catastrophe (Casey directive)
- feedback_limits_lossy.md: limits are lossy compression
- feedback_caseys_principle.md: entropy = force = counting; Gödel = boundary = definition
- T2476: α^{BST primary} substrate-coordinate count framework
- Casey's Curvature Principle: "you can't linearize curvature" (P≠NP via Gauss-Bonnet)

Match precision

TARGET-PREDICTION + NOVEL framework. SP-30-9 is the newest SP-30 sub-item; theoretical foundation requires further development per multi-month framework refinement.

Cal #21 dual-gate status

- EMPIRICAL gate: OPEN (experiments not yet performed)
- MECHANISM gate: PARTIAL (γ as trajectory + catastrophe framework articulated; substrate-mechanism multi-month per K52a Sessions 6+)

Cal #50 DOUBLE-LOCKED EXTERNAL discipline

External register uses operational language only: - External: "BST predicts a class of small modulation signatures in precision spectroscopy near algebraic/transcendental boundaries. Statistical analysis of Rydberg-state series limits + nuclear decay branching can test this prediction." - Internal (this document): γ-trajectory + catastrophe theory + substrate-cognition framework

Cal #99 META-theorem framing

SP-30-9 trajectory spectroscopy is a SUBSTRATE-DERIVATION CONSEQUENCE of: - T2476 α^{BST primary} substrate-coordinate count - Casey's γ-trajectory +

catastrophe framework (memory feedback) - Casey's limits-are-lossy framework (memory feedback)

NOT a new Strong-Uniqueness criterion.

Bibliography

1. Toy 1157 (Elie + Casey 2026 ~April): γ_EM trajectory classification — limit-undecidable numbers.
2. Casey's catastrophe theory connection (memory: feedback_gamma_trajectory.md).
3. Casey's limits-are-lossy framework (memory: feedback_limits_lossy.md).
4. T2476 (Lyra Friday 2026-05-22): α^{BST primary} substrate-coordinate count.
5. Casey's Curvature Principle (memory: feedback_curvature_principle.md): P≠NP via Gauss-Bonnet.

— Elie, SP-30-9 v0.1 paper-grade experimental proposal, 2026-05-23 Saturday 16:38 EDT (`date`-verified; NEW Elie-primary lane; multi-month theoretical refinement + experimental design alongside)